

Juan Pablo Ramirez

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EDUCATION

B.Sc., Mechanical and Aerospace Engineering
Minor in Physics and Mathematics, GPA: 3.427
New Mexico State University (NMSU)

August 2022-December 2026
Las Cruces, NM

WORK EXPERIENCE

• Student Aid at TRIO STEM-H

Tutoring mathematics and engineering mentoring for students in need, as well as operating and maintaining the program's website.

May 2023-Ongoing
Las Cruces, NM

• Southwest Cheese Engineering Intern

Troubleshooted flaws in the cheese production process. Solved problems within an engineering environment and designed systems for human-machine interactions.

May 2026- August 2026
Clovis, NM

• APS BioGroup Engineering and Maintenance Intern

Supported colostrum powder production and manufacturing plant expansion. Managed projects involving Computer Aided Design, technical documentation, and Capital Expenditure.

May 2025- August 2025
Phoenix, AZ

PROJECTS

• Lockheed Martin CommNet Proposal Project

Developed a communication satellite network proposal for competition as a capstone project. Built a prototype for a 3U CubeSat alongside its associated research and simulations. Produced and implemented accurate risk assessment in space mission design.

September 2025-May 2026

• Interceptor Missile Ordinance System

Member of the American Institute of Aeronautics and Astronautics, designed and documented the ordinance system of a missile as part of a design competition.

August 2024-May 2025

• INCA CubeSat Development

Member of the NMSU Nanosatellite Lab structures team. Designed the antenna release system for the neutron-detecting satellite using SolidWorks.

October 2022-May 2025

• Web Developing and Personal Cloud Storage System

Built several websites for different purposes ranging from entertainment to product showcase, among which a storage system developed on Python and hosted on a Linux server.

• Orbit Determination and Observation Program and UI Design

Developed an accessible program to predict, visualize, and animate the movement of planets and satellites in both Heliocentric and Geocentric models, compatible with the Ansys STK Astrogator environment.

• Aircraft Development Workflow Tools

Optimized and streamlined rough aircraft design processes using Python and MATLAB scripts, emphasizing fluid analysis in conditions including cruise flight, orbital re-entry, and wind tunnel simulations.

SOFTWARE AND SKILLS

CAD: SolidWorks, Autodesk Fusion, AutoCAD, FreeCAD, Ansys STK

Programming: MATLAB, MATLAB App Designer, Linux, Python, HTML, CSS, JavaScript, Godot

Media: Blender, OBS, Photoshop, Adobe After Effects

Productivity: MS Office, Google Workspace, Libre Office

Certifications: CSWA, CPR & First Aid, QPR, CRLA. Fluent in Spanish and Portuguese